

V Muhammed Saad Sabeel

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EDUCATION

BMS Institute of Technology & Management

Bachelor of Engineering in Electronics & Telecommunication Engineering, CGPA: 9.21

Bengaluru, India

2023 – 2027

Aragami PU College

Class 12 Science, Percentage: 89%

Bengaluru, India

2022 – 2023

PROJECTS

EVSE Charge Controller Simulator | *ESP32, FreeRTOS, Embedded C*

- Developed embedded firmware in C on ESP32 using FreeRTOS to implement an EV charging state machine.
- Managed concurrent sensor monitoring, PWM generation, interrupt handling, and safety fault detection using tasks, queues, and binary semaphores.

QNX Deterministic Security Monitor | *C, QNX RTOS, Microkernel Architecture, IPC*

- Designed a security monitor on a QNX Microkernel using strictly isolated C programs to mitigate hardware side-channel attacks.
- Developed a high-priority mitigation engine triggered by IPCs to instantly freeze compromised processes mid-instruction and prevent data theft.

Smart Switching Using IR Remote and Mobile | *NodeMCU ESP32, IoT, Blynk*

- Developed a home automation system using a 4-channel relay module for appliance control.
- Implemented dual control modes: an IR remote for local control and the Blynk IoT platform for mobile access.

Diabreeze: Non-Invasive Biomarker Detection System | *ESP32, Data Processing*

- Built a portable diagnostic system that processes real-time breath acetone telemetry to estimate blood glucose levels.
- Engineered an ESP32 data pipeline to instantly compute and visualize physiological and environmental metrics.

IoT Intrusion Detection via Side-Channel Analysis | *Raspberry Pi, I2C, Python, Edge AI*

- Building a non-intrusive edge monitoring system that uses hardware power analysis to detect malware, bypassing the need for software logs.
- Created an I2C data pipeline to sample current consumption every 0.2 seconds, building a dataset to train real-time Edge AI classification models.

EXTRACURRICULAR

AI Model Evaluator | *Outlier AI*

- Contributed 330+ hours of RLHF and SFT tasks to train enterprise LLMs, evaluating model outputs for factual accuracy and logical consistency.
- Performed quality evaluation of AI-generated outputs across Mathematics and multilingual domains.

Webmaster | *IEEE Circuits and Systems Society (CASS)*

- Managed and maintained the IEEE CASS student chapter website, ensuring timely updates of announcements, events, and technical content.
- Collaborated with the organizing team to design and publish event pages.

TECHNICAL SKILLS

Languages: Embedded C, Python

Embedded & OS: ESP32, Raspberry Pi, FreeRTOS, QNX Neutrino Microkernel

Tools & Software: Keil uVision, Git, VS Code, Xilinx Vivado